

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce

Department of Information Technology (B.Sc.I.T Semester V)
MERN STACK

Practical – II

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Class: TYIT	Batch:1
Date of Assignment:30-06-2026	Date/Time of Submission: 30-06-2026

2. Core JavaScript Concepts for MERN

a. Create functions and call them with parameters.

Code :

```
function calSalary(name,basicSalary){  
let bonus = 5000;  
let totalSalary = basicSalary + bonus;  
  
console.log("Name : ",name);  
console.log("Basic Salary : ",basicSalary);  
console.log("Bonus : ",bonus);  
console.log("Total salary : ",totalSalary);  
}
```

```
calSalary("Rajjj", 100000);
```

Code A2:

```
function BillGenerator(product , quantity , price){  
let GST = price * 0.18;  
let totalBill = (price * quantity) + GST;  
  
console.log("Product : ",product);  
console.log("Quantity : ",quantity);  
console.log("Price : ",price);  
console.log("Total bill : ",totalBill);  
console.log("Prajapati raj");  
}
```

```
BillGenerator("Soup", 100, 15000);
```

Output:

```
PS R:\MERN STACK> node prac2a.js
Name : Rajjj
Basic Salary : 100000
Bonus : 5000
Total salary : 105000
```

A2

```
PS R:\MERN STACK> node prac2a.js
Product : Soup
Quantity : 100
Price : 15000
Total bill : 1502700
Prajapati raj
```

b. Work with arrays and objects in JavaScript.

Code :

```
let fruits = ["apple", "banana", "orange", "grapes", "mango"];
let student = {
  name: "Raj",
  age: 20,
  grade: "A",
  mobile: "1234567890"
}
console.log("Fruits: ", fruits);
console.log("Fruits : ", fruits[0]);
console.log("Name : ", student.name);
console.log("Age : ", student.age);
console.log("Grade : ", student.grade);
console.log("Mobile : ", student.mobile);
```

Output:

```
Fruits: [ 'apple', 'banana', 'orange', 'grapes', 'mango' ]
Fruits : apple
Name : Raj
Age : 20
Grade : A
Mobile : 1234567890
PS R:\MERN STACK> 
```

c. Demonstrate use of arrow functions.

Code :

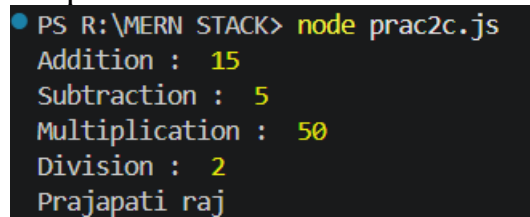
```
let add = (a, b) => {
  return a + b;
};
let sub = (a, b) => {
  return a - b;
};
let mul = (a, b) => {
```

```

    return a * b;
};
let div = (a, b) => {
    return a / b;
};
console.log("Addition : ", add(10, 5));
console.log("Subtraction : ", sub(10, 5));
console.log("Multiplication : ", mul(10, 5));
console.log("Division : ", div(10, 5));
console.log("Prajapati raj");

```

Output:



```

PS R:\MERN STACK> node prac2c.js
Addition : 15
Subtraction : 5
Multiplication : 50
Division : 2
Prajapati raj

```

d. Write a program using array methods (map, filter, reduce).

Code :

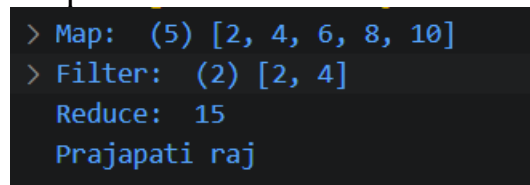
```
let numbers = [1, 2, 3, 4, 5];
```

```
let doubled = numbers.map(x => x * 2);
console.log("Map: ", doubled);
```

```
let evens = numbers.filter(x => x % 2 === 0);
console.log("Filter: ", evens);
```

```
let sum = numbers.reduce((sum, num) => sum + num, 0);
console.log("Reduce: ", sum);
console.log("Prajapati raj");
```

Output:



```

> Map: (5) [2, 4, 6, 8, 10]
> Filter: (2) [2, 4]
Reduce: 15
Prajapati raj

```

e. Create a program that manipulates objects and displays output.

Code :

```
let person = {
    name: "Rajj",
    marks: 85
};
console.log("Name: ", person.name);
console.log("Marks: ", person.marks);
// Manipulate the object

```

```

console.log("Updating marks...");
person.marks = 90;
console.log("Marks: ", person.marks);
console.log("Adding new property country...");
person.country = "India";
console.log("Country: ", person.country);
delete person.marks;
console.log("deleting marks");
console.log(person)

```

Output:

```

C:\Program Files\nodejs\node.exe .\prac2e.js
Name:  Rajj
Marks:  85
Updating marks...
Marks:  90
Adding new property country...
Country:  India
deleting marks
> {name: 'Rajj', country: 'India'}

```

f. Implement a small JavaScript program combining functions, arrays, and objects.

Code :

```

let students = [
    { name: "rajj", age: 20, grade: "A" },
    { name: "pratik", age: 22, grade: "B" },
    { name: "som", age: 21, grade: "C" }
];
function displayStudents(students) {
    console.log("Students details: ");
    console.log("-----");
    for (let stud of students) {
        console.log(`Name : ${stud.name}, Age : ${stud.age}, Grade :
${stud.grade}`);
    }
}
displayStudents(students);

```

Output:

```

C:\Program Files\nodejs\node.exe .\prac2f.js
Students details:
-----
Name : rajj, Age : 20, Grade : A
Name : pratik, Age : 22, Grade : B
Name : som, Age : 21, Grade : C

```